JAFFEDRONE

A drone instrument and meditation tool for the TrimUI Brick and Brick Pro

User manual · version 0.7.0



github.com/sydilaxe/Trimui-Jaffedrone-dev

Getting started



Jaffedrone is a drone of five voices: strings that are bowed and plucked, or wind voices that are blown and tongued, tuned to a scale and moving through chords. Each voice has its own rhythm, or you play it by hand.

Install

Jaffedrone runs on MustardOS. Copy Jaffedrone-mustardos-brick.muxapp (TrimUI Brick and Brick Hammer) or Jaffedrone-mustardos-brickpro.muxapp (Brick Pro: it adds the sticks) into the ARCHIVE folder of the first SD card, install it from Applications → Archive Manager and open Applications → Jaffedrone. The start-up screen shows for a moment; any button moves on.

Updates

Jaffedrone updates itself. A few seconds after it starts it looks for a newer release (over Wi-Fi) and, if there is one, asks: A downloads it and restarts, B asks again next time. The sound goes on meanwhile. Your song, presets, MIDI map and settings are kept, and a failed update changes nothing. While it is open, Jaffedrone also keeps MustardOS from dimming, muting or sleeping when the buttons go quiet, so a drone plays on untouched.

The first buttons

1	START	start or stop the drone (it fades in and out): its rings, chord changes and clock. Only START (or a MIDI or Wi-Fi START) starts it; playing by hand or from MIDI sounds with it stopped
2	L1 / R1	previous / next page: PLAY, SOUND, COMPOSE, MIX, SONG, MIDI
	SELECT	the next chord
3	MENU	leave the app

Listen on headphones for the deepest low end and for binaural beats; the speaker plays little below about 200 Hz.

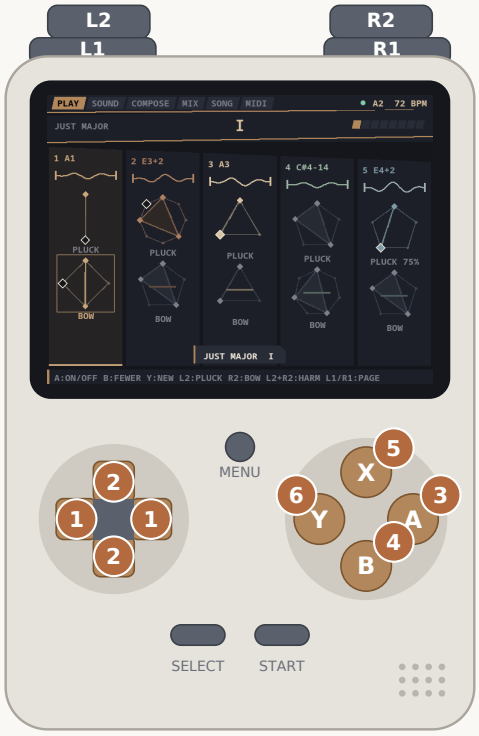
The six pages



PLAY: the strings and their rhythms | SOUND: each voice's tone | COMPOSE: scale, chords, change | MIX: levels and effects | SONG: presets, chord memory, key | MIDI: channels, keys, clock, CC map

R1 moves right along the tabs at the top, L1 left, on every page (round from MIDI to PLAY).

PLAY: the rhythm rings

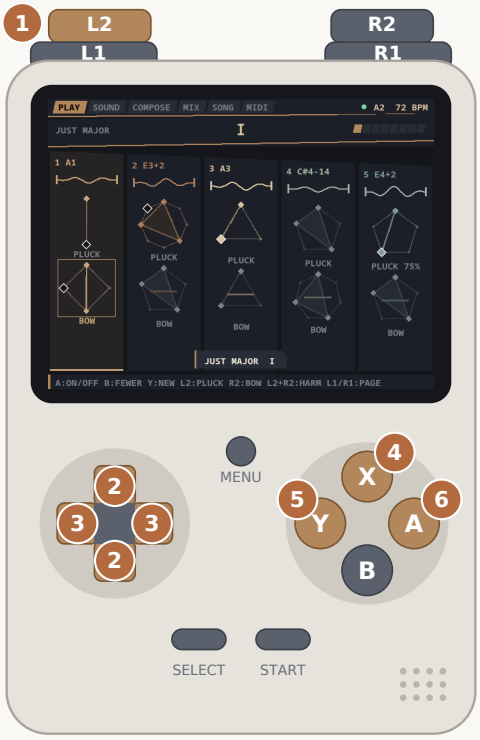


Each string has two Euclidean rings: the pluck ring plucks on its pulses and the bow ring swells the bow on its pulses (off, the bow plays steadily). A ring spreads its pulses as evenly as it can over its steps, like the rhythms of HI5 and RadioScan. Each string keeps its own step length, so they play polyrhythms: eighths against eighth triplets, sixteenths against 5/16ths. ACCENT (MIX) makes a string's hits stand out of the drone.

1	◄►	choose a string
2	▲▼	choose its pluck ring or bow ring
3	A	switch the ring on or off
4	B	fewer pulses, wrapping back to all
5	X	rotate the pattern
6	Y	a new simple pattern

A label like PLUCK 70% means each pulse of that ring plays seven times in ten (set with R2 + X / Y).

Shaping a ring: hold L2



1	hold L2	then:
2	▲▼	more / fewer steps
3	◄►	step length (right is faster): 1/32 to 2 bars, with triplets (1/16T, 1/8T, 1/4T), dotted (1/8., 1/4.) and quintuple (5/16, 5/8)
4	X	more pulses
5	Y	rotate back
6	A	mute or unmute the string (a red bar marks it)

Playing by hand

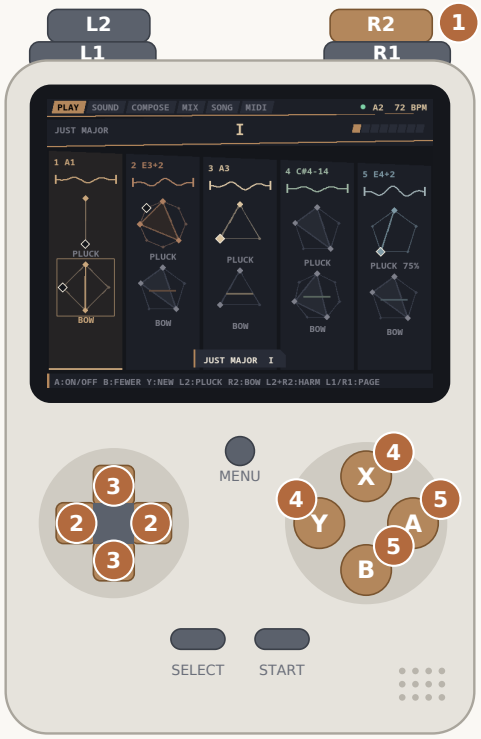
The lower shoulder buttons play the string selected on PLAY, over whatever its rings are doing. A wind voice is tongued and blown instead.



1	L2	pluck the selected string, the moment you press (wind: tongue a note; struck: the mallet). Held, L2 still shapes the ring
2	START + L2	drop a dulcimer mallet on it: a bright hit, then the mallet bounces, each rebound lower and sooner, until it rests (wind: a flutter tongue, the breath rolling "frrr" and settling; struck: a hard mallet, bouncing)
3	R2 held	bow it: the bow is drawn in slowly, swelling over about two seconds and brightening as it bites, and sings while you hold (wind: blow). Press another button with R2 held and it is a quick sound change (below): the bow lifts
	L2 + R2, either first	bow with a light touch at a node: the next harmonic up sings (wind: overblow, an octave up, a twelfth for the reed). Bowing with R2, add L2 for overblows as you bow

The bowed string's column lights while the bow is on it.

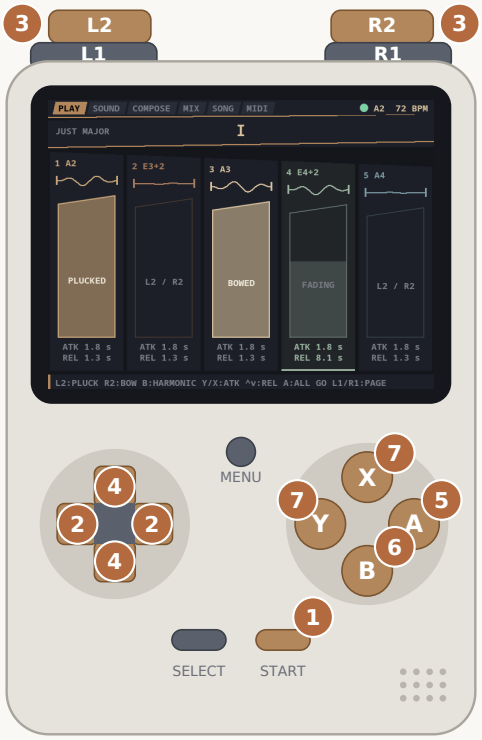
Quick sound changes: hold R2



1	hold R2	then:
2	◄ ►	previous / next scale
3	▲ ▼	tone: brighter / darker
4	X / Y	the selected ring's chance, more / less
5	A / B	body resonance, more / less

LATCH: hold each string yourself

In LATCH the rings rest and every string is silent until you play it. A string you pluck or bow keeps droning, with no need to hold the button, until you play it again; then it fades over its own RELEASE. The drone needn't be started; stopping it (START) lets every latched string go.



1	START + A	switch PLAY MODE between RINGS and LATCH (also on the SONG page)
2	◄►	choose a string
3	L2 / R2	pluck / bow it on: it drones (START + L2: the mallet drops it on). Again: it lets go and fades. R2 held for a quick sound change puts the string back as it was
4	▲▼	that string's RELEASE, 0.05 s to 20 s
5	A	let every string go
6	B held, or L2 + R2	the next harmonic up on the droning string (wind: overblow); with both shoulders, either first, it lasts while both are held
7	Y / X	that string's bow ATTACK, shorter / longer, 0.05 s to 10 s

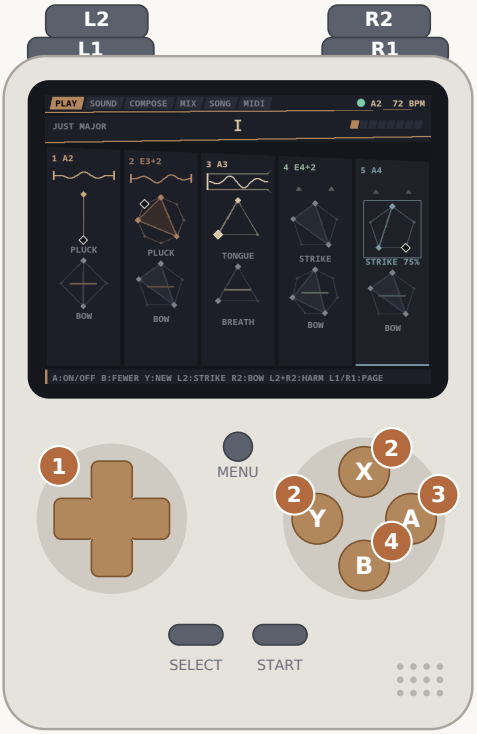
Each column shows a board with a shed-roof top. It fills while its string drones and drains as it fades, marked STRUCK, PLUCKED, BOWED, HARMONIC or FADING, with its ATTACK and RELEASE beneath. A pluck or strike starts at once; a bow swells in over its ATTACK. RELEASE is also on the MIX page; LATCH, ATTACK and RELEASE are saved with songs and presets.

Try it

Press START + A. Bow string 1 with R2 for a low drone, move to string 3 and pluck it with L2, then press L2 on string 3 again to let it ring away. Raise string 5's RELEASE with ▲ to ten seconds or more for a long dying shimmer. Press START + L2 on string 4 to drop the mallet on it, and hold B to let its harmonic sing. Lower string 1's ATTACK with Y for a bow that bites at once.

SOUND, COMPOSE and MIX

On these pages the d-pad moves to the nearest control in that direction, and Y / X lower / raise it. Held, they speed up.



1	d-pad	move between controls
2	Y / X	lower / raise the control
3	A	on a string's control: mute or unmute that string
4	B	pluck the string (to hear a change)

SOUND

The VOICE row makes each column a STRING, one of seven winds (LIP, REED, DOUBLE, FLUTE, FREE, RECORDER, BANSURI) or one of nine struck voices (see Struck voices). R2 + Y / X on VOICE jumps to the previous / next family; Y / X alone step one at a time. The rows below follow the voice: a string has BOW, DECAY, BRIGHT, FOLD, STIFF, POS and BUZZ; a wind BREATH, AIR, EMBOUCH, VIBRATO and GROWL; a struck voice BOW, RING, MALLET, STRETCH, SPOT, SIZZLE and SING. The shared row holds COUPLE, BODY, its TYPE, the winds' BORE and CUTOFF; the last row the sympathetic strings.



COMPOSE: the scale and root (a wheel of the scale's notes, the chord lit), how often chords change and how: MOTION, HOME, COLOR, CHANCE, RELATE, EVOLVE, GLIDE, tempo, Wi-Fi sync and two LFOs.



MIX: each voice's LEVEL, PAN, GATE (RELEASE in LATCH), ACCENT (how its pluck ring's hits stand out: the bow lifts, the hit is louder) and AURA, the master, undertones (SUB), then AURA, delay, reverb and binaural.

AURA: voices that listen

AURA lets some voices sound only from the others, as if they stood beside them in one room: a gong humming along with a flute, bowls singing when a reed plays. On MIX, set each voice's AURA to SEND (it plays, and its sound goes into the air), LISTEN (it is never bowed, blown, struck or plucked; it sounds only from what reaches it) or OFF (it plays, out of the air). Then raise AMOUNT under AURA, from OFF to 100.

Each voice stands where its PAN puts it: the players' sound reaches a listener later and fainter the further it stands. A listener rings where its own notes meet the players': struck bodies with a long RING sing most. The sound only goes one way, from players to listeners (and from one listener to the next), so it can never feed back. With every voice on SEND, AURA does nothing: at least one must listen. Delay, reverb and binaural come after, on everything.

Struck voices

A struck voice is a physical model of a vibrating body: its resonant modes, struck by a mallet (a mass on a spring that meets the surface and bounces off, so a hard mallet makes a short bright contact and a soft one a long dark push) or bowed (stick-slip friction, so the modes it meets grow until they sing, as on a rubbed bowl or a bowed cymbal). Every voice can be struck and bowed.

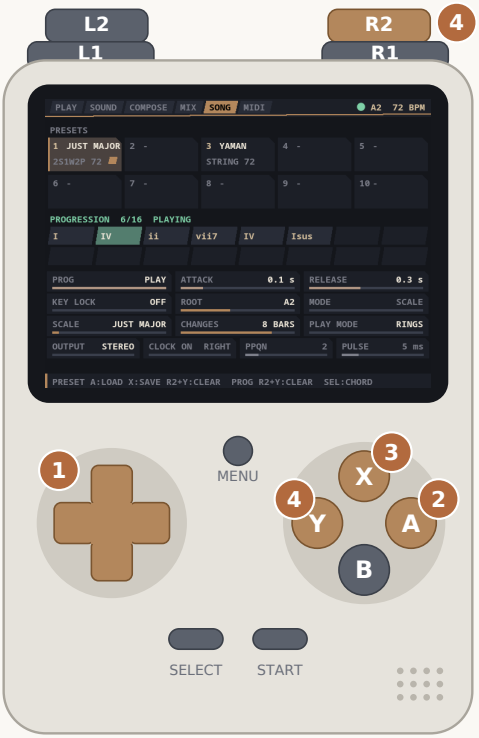


1	L2	strike (START + L2: a hard mallet, dropped, bouncing)
2	R2	bow: the body catches and sings
3	L2 + R2, B	a touch: the fundamental is damped and the next mode rings and sings

GENDER	gamelan metallophone bars, in beating pairs (ombak)
BONANG	gamelan kettle gongs, paired
GONG	a large gong, low, its pairs beating slowly
BOWL	a singing bowl: struck, or rubbed until it sings
HANDPAN	fundamental, octave and fifth above, tuned
STEELPAN	the same family, brighter
TONGUE	a steel tongue drum: pure, a bar held at one end
CYMBAL	a dense shimmer of modes; bowed, it screams and sings
DRUM	a membrane: a short pitched thud
GLASS	a glass harmonica: pure, long, made to be bowed
CHORD	metal tuned to the scale: it rings as a chord
ALIEN	golden-ratio modes: a metal no body has
RING	how long it rings; from 95 up a metal drone (DRONE)
MALLET	soft felt to hard metal
STRETCH	pulls the modes apart, far at the top: unearthly clangor
SPOT	where it is struck and bowed: which modes speak
SIZZLE	a rattle that follows the ring
SING	the mode the bow favours
BOW	bow pressure: from a whisper to a scream

Each struck body has its own limiter, so strikes on a long ring never pile up. The mode frequencies follow measurements and theory: rectangular-bar modes and the makers' harmonic partials for the gamelan bars (Sethares), 1 : 2 : 3 for handpan and pan, 1, 2.9, 5.7, 9.1 for the bowl (Inacio, Henrique and Antunes), a clamped bar for the tongue drum, a thin plate's dense series for the cymbal, the Bessel-function modes of a membrane for the drum.

SONG: presets, chord memory and key



1	d-pad	choose a preset slot or a setting
2	A	load the preset (your Wi-Fi and output settings stay)
3	X	save the sound and rhythms to the slot
4	R2 + Y	clear the slot, or the chord memory

Ten preset slots keep everything: voices, rhythms, scale, effects, the chord memory and PLAY MODE.

Chord memory

Set PROG to RECORD and every chord change (SELECT or CHANGES) is kept, up to 16. PLAY steps through them round and round; FREE lets the algorithm choose again.

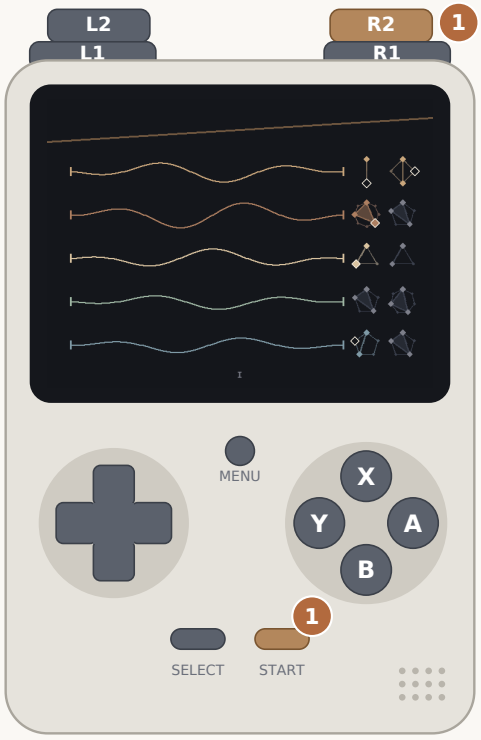
Playing with others

KEY LOCK holds the root, scale and MODE (the mode chords come from in EQUAL) and tunes chords exactly. ATTACK and RELEASE set how the drone fades in and out with START.

Output: mono + clock

OUTPUT MONO+CLK plays the sound in mono on one side of the headphone jack and clock pulses on the other, to lead a Pocket Operator, Volca or other gear with a sync input (use a stereo-to-two-mono cable). CLOCK ON picks the side, PPQN the pulses per quarter note (2 for Pocket Operators and Volcas, 24 for DIN sync) and PULSE the width. The pulses run while the drone plays, on the beat and with Wi-Fi sync. Every PAN and stereo spread plays from the centre; the settings are kept for STEREO, and binaural becomes OMBAK.

The quiet view



START + R2 on PLAY for the quiet view: the five strings across the screen with their rings, and only the chord in words. Again to come back.

1	START + R2	the quiet view on / off

Binaural, ombak and tunings

On MIX, BEAT sets a binaural beat from 0.1 to 40 Hz: the voices below about 1 kHz are shifted down in the left ear and up in the right, so every low partial beats between the ears. DEPTH sets how much. MODE:

FREE	the beat as set
TEMPO	the beat moves to a power-of-two multiple of the tempo
A>T 10M	a journey from alpha (10 Hz) down to theta (6 Hz) over ten minutes
T>D 15M	from theta (6 Hz) down to delta (2.5 Hz) over fifteen minutes
OMBAK	the beat for both ears, like a Balinese gamelan's paired instruments: the low end swells and fades, and it works on the speaker

Settings to start from

Meditation	BEAT 6 MEDITATE, headphones; or MODE A>T 10M to drift down from alpha
Sleep	MODE T>D 15M, headphones, a long ATTACK and RELEASE
Gamelan shimmer	scale TIRUS or BEGBEG, MODE OMBAK, BEAT 7 Hz, on the speaker

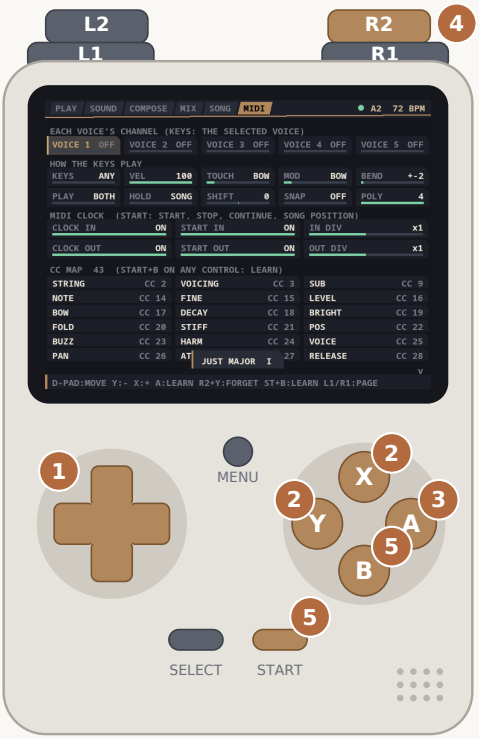
The research on binaural beats is limited and mixed; these are the settings most used, not a promise of an effect.

Tunings

COMPOSE, or R2 + ◄ / ► on PLAY, chooses the scale: EQUAL (with its modes), just major and minor, the harmonic series, just pentatonics, the ragas YAMAN, BHAIRAV, KAFI, BHUPALI and MALKAUNS, the gamelan tunings SLENDRO, PELOG and SELISIR, and the Balinese gong kebyar tunings BEGBEG, SEDANG and TIRUS (Toth's profiles, as reproduced by Vitale and Sethares). Each scale has its own way of moving through chords.

The MIDI page

A MIDI keyboard or controller plays the voices and moves the settings. The MIDI page sets how; it belongs to the Brick, not the song (kept in midi.cfg, kept when a preset loads).



1	◄►▲▼	choose a setting or a CC
2	Y / X	change it (on the map: the CC, by hand)
3	A	on the map: learn the entry's CC again (move a knob)
4	R2 + Y	on the map: forget the entry
5	START + B	MIDI learn, on any page's selected control; again to cancel

VOICE 1-5	each voice's own channel (OFF, or 1-16): its keys, and its CCs, come on it
KEYS	the channel of the keys that play the voice selected on PLAY: ANY (every channel that is no voice's) or one
VEL	how much velocity matters: 100 as struck, 0 every key alike
TOUCH	what aftertouch moves, a key's own on its note only: BOW (harder), TONE (brighter), SWELL (level, from 35% up), VIBRATO, PITCH (up to two semitones), HARM (the next harmonic past halfway), FOLD, BUZZ, or OFF
MOD	what the mod wheel moves, the same choice
BEND	the pitch bend range, 1 to 24 semitones
PLAY	what a key does: BOTH plucks and bows while held, PLUCK only plucks (it rings on), BOW only bows (it swells in), STRIKE drops the mallet
HOLD	how long: SONG as PLAY MODE (RINGS while held, LATCH until let go), GATE while held, LATCH a key drones on and the same key again lets it go
SHIFT	every key moved, -24 to +24 semitones
SNAP	SCALE: each key moves to the nearest note of the song's scale and root
POLY	how many keys a voice plays at once, 1 to 4 (at 1, each key takes the last's place)
CLOCK IN	MIDI CLOCK: ON follows a clock that comes in; OFF ignores it
CLOCK OUT	ON sends clock while the drone plays (with CLOCK IN on, until one comes in); OFF sends none
START IN	ON starts, stops and places the drone with the clock's START, STOP, CONTINUE and song position; OFF keeps the clock's tempo only, so you start and stop it
START OUT	ON sends START and STOP with the clock; OFF sends the clock alone
IN DIV	the drone at a ratio of the clock coming in, 1/4 to x4 (1/2: half time)

OUT DIV	the clock sent at a ratio of the drone's tempo, 1/4 to x4 (x2: double time)
CC MAP	each CC and what it sets. Select any control on SOUND, COMPOSE, MIX or SONG, press START + B and move a knob: that CC now sets it. A voice's control moves the voice of the channel the CC comes on (the selected one on the others). CCs 1, 64, 74, 91, 120 and 123 are fixed

Each key sounds its voice (its instrument and SOUND settings, level and pan) at its own pitch on one of eight key voices beside the strings, so the drone and the song are untouched and a muted string still plays from the keys. A voice plays four keys at once; a fifth takes the place of its key held longest.

USB: MIDI and audio

The Brick's USB-C port takes class-compliant MIDI and audio devices (through an OTG adapter or a hub): keyboards, controllers, sequencers, drum machines and USB sound cards. They are found as they are plugged in; a message names each one.

MIDI in

Notes	a key plays the voice of its channel (the MIDI page), at its own pitch, four at once a voice; it plucks as hard as it is struck and bows while held (a wind voice is tongued and blown). In LATCH a key drones (a soft key swells in with the bow, a firm one plucks) until let go, then fades over the voice's RELEASE. Keys sound with the drone stopped and never start it
Pitch bend	BEND semitones (2 at first)
Aftertouch, mod wheel	what TOUCH and MOD move (BOW at first)
CC64	sustain
CC74	TONE: darker / brighter
CC91	more / less reverb
Clock	the tempo follows it (fitted over the last four beats of ticks, so USB's jitter averages out), its phase held on the sound as it is heard; START, CONTINUE, STOP and song position start, stop and place the drone

MIDI out

While the drone plays, and no clock is coming in, Jaffedrone sends MIDI clock (24 a quarter note) with START and STOP to every MIDI port, timed to the sound as it is heard (the MIDI CLOCK switches, below, turn each off; OUT DIV sends it faster or slower). It leads a drum machine or sequencer over USB as MONO+CLK does over the headphone jack.

MIDI CLOCK (on the MIDI page)

CLOCK IN	ON (the default) follows a clock that comes in; OFF ignores it (notes and CCs still come in)
CLOCK OUT	ON (the default) sends clock while the drone plays; with CLOCK IN on, only until a clock comes in. OFF sends none
START IN	ON (the default) starts, stops and places the drone with the clock's START, STOP, CONTINUE and song position; OFF follows the clock's tempo only, so you start and stop it yourself
START OUT	ON (the default) sends START and STOP with the clock; OFF sends the clock alone, for gear you start by hand
IN DIV	the drone's tempo as a ratio of the clock that comes in: 1/4, 1/3, 1/2, 2/3, x1 (the default), 3/2, x2, x3, x4. At 1/2 a drum machine at 120 drives the drone at 60, half time; song position lands at the same ratio
OUT DIV	the clock sent as a ratio of the drone's tempo, the same choice: at x2 the gear runs double time

They belong to the Brick, like WIFI SYNC: loading a preset keeps them.

Linking Bricks

Several Bricks play in time with one by cable, with no MIDI interface: the leader has a powered USB hub in its top port, and each follower plugs its bottom (charging) port into the hub with a data cable. The wiring decides who leads. Nothing to set up: Jaffedrone offers its bottom port as a "TrimUI clock link", and a leader sends its clock, START and STOP and where its sound is, about 50 times a second, so the followers sound together within a tenth of a millisecond, up to 250 BPM. A follower plugged in mid-song comes in at the leader's place; one with START IN off keeps the leader's tempo and bar but starts and stops by hand.

Columbia Street, HI5 and RadioScan link the same way, so any of them can lead. A leader can use USB audio and MIDI gear on the same hub; one that follows a drum machine's clock passes it on. JD_LINK=0 in launch.sh leaves the bottom port alone.

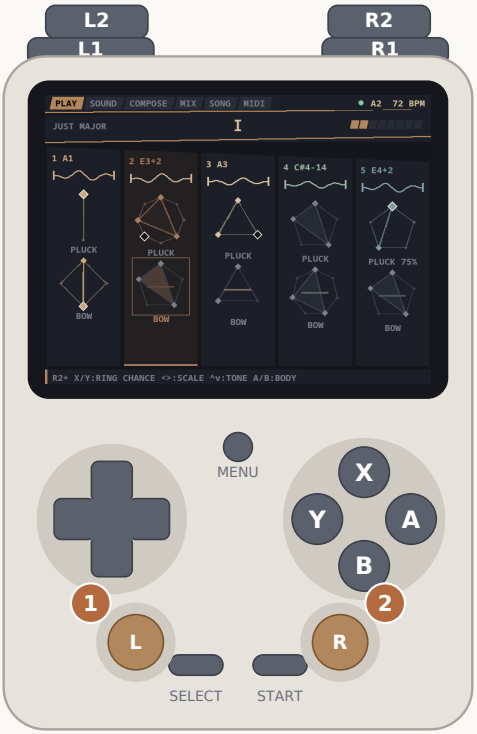
Audio

A USB sound card or DAC, once plugged in, takes the sound from the speaker and headphones; unplugged, the sound comes back to them.

Settings

In launch.sh in the app folder: JD_MIDI_NOTES=CHANNELS gives voices 1-5 channels 1-5 until the MIDI page sets them; JD_MIDI_CLOCK=0 never sends clock; JD_MIDI_FOLLOW=0 never follows it, whatever MIDI is set to; JD_AUDIO=internal keeps the sound on the speaker and headphones.

Brick Pro: the sticks



On the Brick Pro the sticks play the string selected on PLAY, and spring back when let go: nothing they do is saved. Each stick reaches its controls' whole range, straight or on a diagonal: it learns how far it really goes the first time it is pushed all the way.

1	left stick ▲	bow the string by hand: a light touch draws the bow lightly, all the way up as hard as the bow goes, half again as hard as R2 (a wind voice: blow)
	left stick ◀ ▶	bend the string, up to two semitones either way
2	right stick ▲ ▼	HARM: up touches higher harmonics, to the 5th; down, lower, to none (a wind voice: its register; struck: the mode the bow sings)
	right stick ◀ ▶	the SOUND control on the stick (BRIGHT at first): right up to its most, left down to its least
	L3 (left click)	pluck the string
	R3 (right click)	the next chord; on SOUND, put the row under the cursor on the right stick

JD_BEND=7 in launch.sh bends up to seven semitones.

Choosing the right stick's control



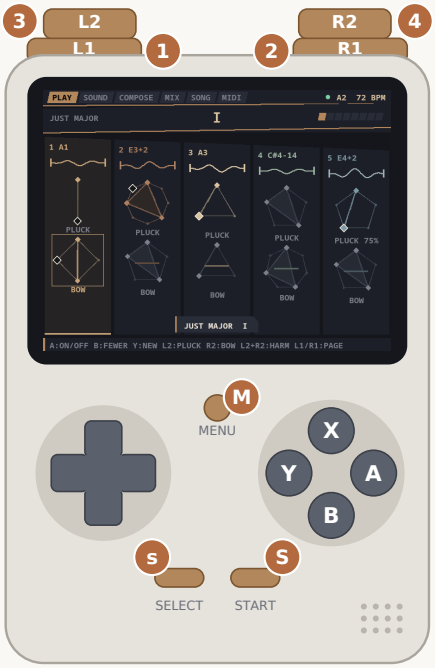
On SOUND, move to a row (BOW, DECAY, BRIGHT, FOLD, STIFF, POS, BUZZ or NOTE) and click the right stick 1: its left / right now plays that control. The choice is kept. The rows the stick plays are marked on the selected string's column, and light up while the stick moves them; the row shows the value playing.

The lights

The lights around the sticks show how far they are pushed: the left glows amber as it bows and rose as it bends, the right green for the harmonics and gold for its SOUND control, brighter the further each goes. At rest they are dark; on leaving, they show what they showed before.

In launch.sh: JD_LEDS=0 leaves the lights alone; JD_STICK_REACH=80 takes a stick to reach 80% of the way before it shows how far it goes (60 at first).

Quick reference



START	start / stop the drone
L1 / R1	previous / next page
START + A	PLAY MODE: RINGS / LATCH
LATCH B / YX	harmonic (held) / bow attack
START + SELECT	the log: OFF / BASIC / DETAIL
START + B	MIDI learn the selected control
SELECT	next chord
L2	pluck (wind: tongue); LATCH: on / let go
START + L2	drop the dulcimer mallet (wind: flutter tongue)
R2	bow while held (wind: blow); LATCH: bow on / let go
L2 + R2	harmonic (wind: overblow), either first
START + R2	quiet view
L2 + **...**	shape the ring, mute (A)
R2 + **...**	scale, tone, chance, body
MENU	leave

The log

START + SELECT cycles the log through OFF, BASIC and DETAIL. It is written to jaffedrone_log.txt in the app folder: how hard the audio is working, any dropouts, the levels, each voice's tuning (DETAIL) and what was played, with suggestions when something could be better. If you hear crackles, set JD_BUFFER=1024 in launch.sh.